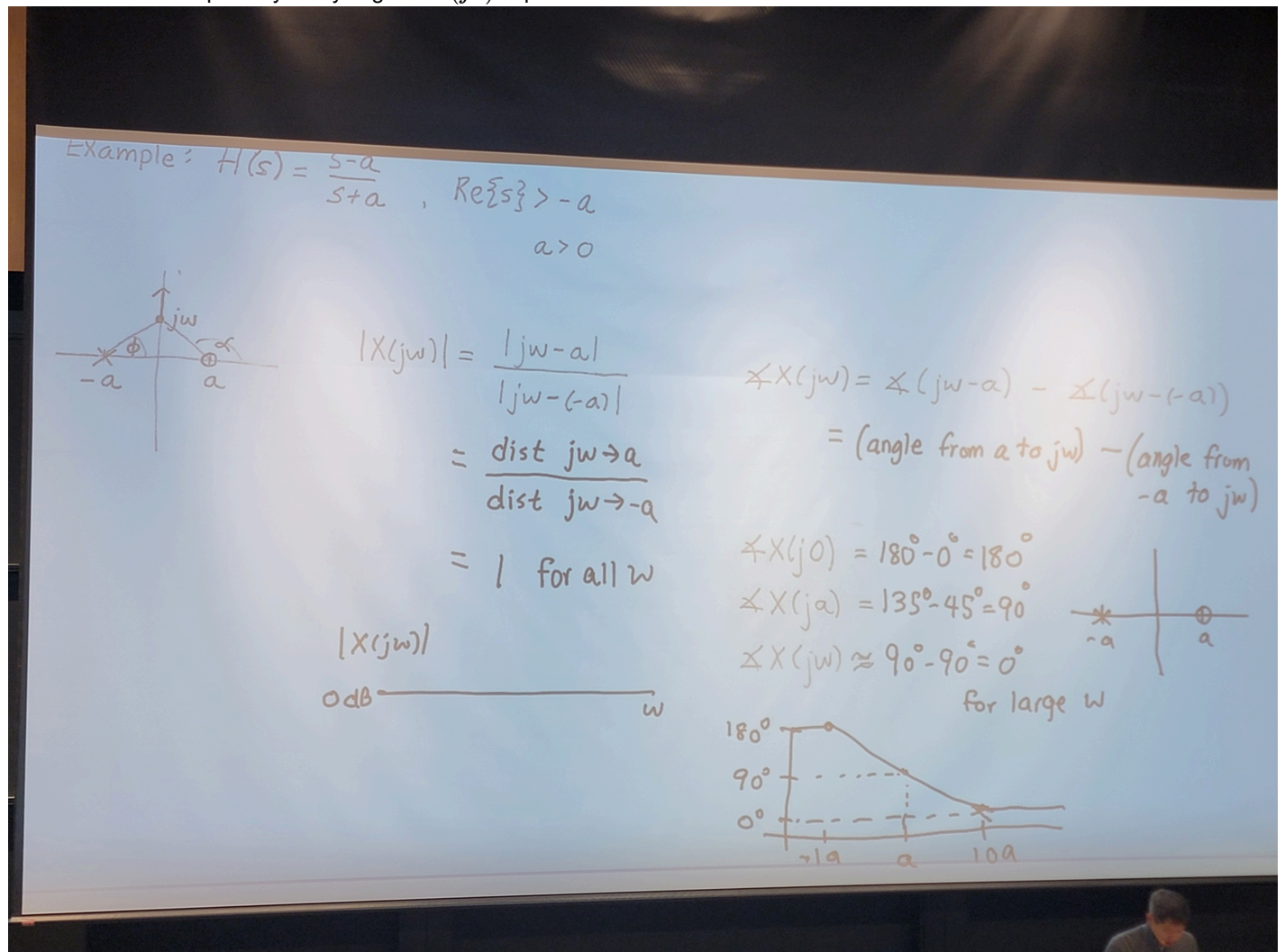
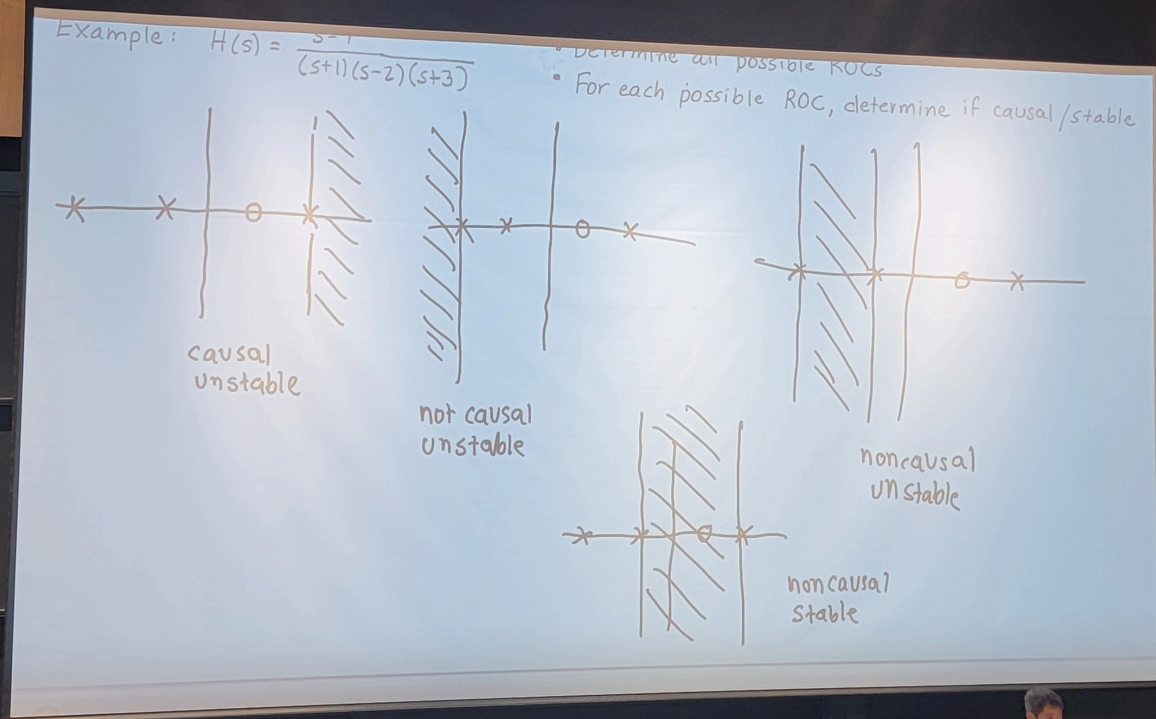


21 Bode Plots

We can make bode plots by analyzing our $X(j\omega)$ expression.



This is huge. This tells us how we can build filters to perform different behaviors. We have zeros as the numerator and poles as the denominator.



If the region of convergence contains the $j\omega$ axis, it is stable, otherwise unstable. If it lies on the right half plane entirely, it is causal. If it involves the left half plane, it is non causal.

Aside: Unilateral Laplace transform

We are assuming that we have a right sided signal where it is 0 before $t = 0$

• LCCDE

LT handles LCCDEs very well.

┌ Aside: Unilateral Laplace transform

$$\mathcal{X}(s) = \int_{0^-}^{\infty} x(t) e^{-st} dt$$

$$x(t) \xleftrightarrow{\mathcal{L}} \mathcal{X}(s)$$

Note: • Unilateral & bilateral LTs are

• useful for LCCDEs with

• helpful property: $x'(t) \xleftrightarrow{\mathcal{L}} s \mathcal{X}(s) - x(0^-)$

$$x''(t) \xleftrightarrow{\mathcal{L}} s(s \mathcal{X}(s) - x(0^-)) - x'(0^-)$$

$$= s^2 \mathcal{X}(s) - s x(0^-) - x'(0^-)$$

└ • ROC must be